

1 Tables for depth $P = 1$

	graph_type depth	ER 1	HH 1	L2F 1	RR 1
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	20 [40–50]	31 [39–144]	130 [40–100]	41 [40–100]
	I_MPS	20 [40–50]	31 [39–144]	130 [40–100]	41 [40–100]
	RTS_MPS	-	1 [106]	-	2 [40–100]
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	20 [40–50]	31 [39–144]	130 [40–100]	41 [40–100]
	I_PP	20 [40–50]	31 [39–144]	130 [40–100]	41 [40–100]
	RTS_PP	-	1 [106]	-	2 [40–100]
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	-	-	-	60 [10–20]
	FA_SV_opt	-	-	-	60 [10–20]
	F_SV	-	-	-	1 [10]
	I_SV	82 [10–20]	10 [12]	276 [10–22]	177 [10–20]
	TQA_SV_noOpt	-	-	-	60 [10–20]
	TQA_SV_opt	-	-	-	60 [10–20]
	TS_SV	81 [10–20]	10 [12]	274 [10–22]	164 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 1	HH 1	L2F 1	RR 1
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	0.8150 ± 0.0118	0.7542 ± 0.0118	0.7267 ± 0.0166	0.7697 ± 0.0153
	I_MPS	0.8150 ± 0.0118	0.7542 ± 0.0118	0.7267 ± 0.0166	0.7697 ± 0.0153
	RTS_MPS	-	0.744	-	0.75170 ± 0.00329
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	0.8150 ± 0.0118	0.7542 ± 0.0118	0.7267 ± 0.0166	0.7697 ± 0.0153
	I_PP	0.8150 ± 0.0118	0.7542 ± 0.0118	0.7267 ± 0.0166	0.7697 ± 0.0153
	RTS_PP	-	0.744	-	0.75170 ± 0.00329
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	-	-	-	0.7801 ± 0.0202
	FA_SV_opt	-	-	-	0.7813 ± 0.0206
	F_SV	-	-	-	0.774
	I_SV	0.7880 ± 0.0291	0.7929 ± 0.0297	0.7427 ± 0.0328	0.8122 ± 0.0432
	TQA_SV_noOpt	-	-	-	0.7807 ± 0.0203
	TQA_SV_opt	-	-	-	0.7813 ± 0.0206
	TS_SV	0.7879 ± 0.0293	0.7929 ± 0.0297	0.7428 ± 0.0327	0.8145 ± 0.0441

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

2 Tables for depth $P = 2$

graph_type depth		ER 2	HH 2	L2F 2	RR 2
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	10 [50]	-	90 [100]	30 [40-70]
	F_MPS	20 [40-50]	31 [39-144]	130 [40-100]	41 [40-100]
	I_MPS	20 [40-50]	31 [39-144]	130 [40-100]	41 [40-100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	20 [40-50]	10 [39]	130 [40-100]	40 [40-70]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	10 [50]	-	90 [100]	30 [40-70]
	F_PP	20 [40-50]	31 [39-144]	130 [40-100]	41 [40-100]
	I_PP	20 [40-50]	31 [39-144]	130 [40-100]	41 [40-100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	20 [40-50]	10 [39]	130 [40-100]	40 [40-70]
SV	FA_SV_noOpt	71 [10-20]	-	195 [10-22]	133 [10-20]
	FA_SV_opt	71 [10-20]	-	195 [10-22]	133 [10-20]
	F_SV	-	-	-	1 [10]
	I_SV	82 [10-20]	10 [12]	276 [10-22]	166 [10-20]
	TQA_SV_noOpt	82 [10-20]	10 [12]	275 [10-22]	133 [10-20]
	TQA_SV_opt	82 [10-20]	10 [12]	274 [10-22]	133 [10-20]
	TS_SV	81 [10-20]	10 [12]	274 [10-22]	132 [10-20]

Table 3: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 2	HH 2	L2F 2	RR 2
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	0.8036 ± 0.0178	-	0.7986 ± 0.0236	0.7567 ± 0.0371
	F_MPS	0.7024 ± 0.0166	0.7213 ± 0.0827	0.7608 ± 0.0427	0.7432 ± 0.0429
	I_MPS	0.7937 ± 0.0134	0.7591 ± 0.0615	0.7923 ± 0.0302	0.7504 ± 0.0423
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	0.7742 ± 0.0131	0.8307 ± 0.0157	0.7923 ± 0.0302	0.7637 ± 0.0358
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	0.87775 ± 0.00994	-	0.7943 ± 0.0175	0.8364 ± 0.0131
	F_PP	0.8595 ± 0.0388	0.8076 ± 0.0217	0.7911 ± 0.0171	0.8038 ± 0.0252
	I_PP	0.8745 ± 0.0115	0.8272 ± 0.0123	0.7957 ± 0.0203	0.8384 ± 0.0137
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	0.8745 ± 0.0116	0.8313 ± 0.0161	0.7957 ± 0.0203	0.8386 ± 0.0138
SV	FA_SV_noOpt	0.8414 ± 0.0215	-	0.8231 ± 0.0287	0.8658 ± 0.0298
	FA_SV_opt	0.8547 ± 0.0231	-	0.8262 ± 0.0290	0.8736 ± 0.0329
	F_SV	-	-	-	0.828
	I_SV	0.8530 ± 0.0243	0.8555 ± 0.0177	0.8200 ± 0.0321	0.8556 ± 0.0509
	TQA_SV_noOpt	0.7611 ± 0.0316	0.7302 ± 0.0503	0.7176 ± 0.0429	0.7598 ± 0.0417
	TQA_SV_opt	0.8549 ± 0.0239	0.8664 ± 0.0247	0.8249 ± 0.0284	0.8736 ± 0.0332
	TS_SV	0.8520 ± 0.0244	0.8708 ± 0.0188	0.8142 ± 0.0281	0.8482 ± 0.0445

Table 4: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

3 Tables for depth $P = 3$

	graph_type depth	ER 3	HH 3	L2F 3	RR 3
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	10 [70]
	F_MPS	10 [40]	31 [39–144]	130 [40–100]	31 [40–100]
	I_MPS	10 [40]	31 [39–144]	130 [40–100]	31 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	10 [40]	10 [70]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	10 [70]
	F_PP	10 [40]	31 [39–144]	130 [40–100]	31 [40–100]
	I_PP	10 [40]	31 [39–144]	130 [40–100]	31 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	10 [40]	10 [70]
SV	FA_SV_noOpt	71 [10–20]	-	177 [10–20]	132 [10–20]
	FA_SV_opt	71 [10–20]	-	177 [10–20]	132 [10–20]
	F_SV	-	-	-	1 [10]
	LSV	82 [10–20]	10 [12]	274 [10–22]	166 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	274 [10–22]	132 [10–20]
	TQA_SV_opt	82 [10–20]	10 [12]	274 [10–22]	132 [10–20]
	TS_SV	81 [10–20]	10 [12]	274 [10–22]	131 [10–20]

Table 5: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 3	HH 3	L2F 3	RR 3
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	0.7640 ± 0.0102
	F_MPS	0.8578 ± 0.0693	0.7558 ± 0.0682	0.8013 ± 0.0439	0.7654 ± 0.0507
	I_MPS	0.8859 ± 0.0197	0.7901 ± 0.0699	0.8567 ± 0.0162	0.8053 ± 0.0716
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	0.8885 ± 0.0108	0.7304 ± 0.0125
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	0.86971 ± 0.00950
	F_PP	0.8794 ± 0.0150	0.8388 ± 0.0272	0.8096 ± 0.0263	0.8466 ± 0.0191
	I_PP	0.9023 ± 0.0135	0.8673 ± 0.0111	0.8327 ± 0.0188	0.8724 ± 0.0149
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	0.7972 ± 0.0240	0.86963 ± 0.00952
SV	FA_SV_noOpt	0.8710 ± 0.0183	-	0.8724 ± 0.0253	0.8966 ± 0.0219
	FA_SV_opt	0.8911 ± 0.0190	-	0.8778 ± 0.0255	0.9105 ± 0.0268
	F_SV	-	-	-	0.874
	I_SV	0.8885 ± 0.0200	0.8976 ± 0.0148	0.8683 ± 0.0265	0.8897 ± 0.0443
	TQA_SV_noOpt	0.8121 ± 0.0385	0.7971 ± 0.0481	0.7627 ± 0.0544	0.8052 ± 0.0507
	TQA_SV_opt	0.8765 ± 0.0220	0.8821 ± 0.0198	0.8450 ± 0.0351	0.8987 ± 0.0261
	TS_SV	0.8797 ± 0.0253	0.9008 ± 0.0182	0.8498 ± 0.0237	0.8741 ± 0.0404

Table 6: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

4 Tables for depth $P = 4$

	graph_type depth	ER 4	HH 4	L2F 4	RR 4
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	30 [100]	10 [40]
	F_MPS	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	I_MPS	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	10 [40]	10 [39]	130 [40–100]	20 [40]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	30 [100]	10 [40]
	F_PP	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	I_PP	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	10 [40]	10 [39]	130 [40–100]	20 [40]
SV	FA_SV_noOpt	52 [10–20]	-	63 [10–20]	73 [10–20]
	FA_SV_opt	52 [10–20]	-	63 [10–20]	73 [10–20]
	F_SV	-	-	-	1 [10]
	LSV	82 [10–20]	10 [12]	274 [10–22]	166 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	274 [10–22]	133 [10–20]
	TQA_SV_opt	82 [10–20]	10 [12]	274 [10–22]	133 [10–20]
	TS_SV	81 [10–20]	10 [12]	274 [10–22]	132 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 4	HH 4	L2F 4	RR 4
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	0.9054 ± 0.0143	0.92585 ± 0.00814
	F_MPS	0.7807 ± 0.0767	0.7595 ± 0.0820	0.8298 ± 0.0437	0.8788 ± 0.0594
	I_MPS	0.9280 ± 0.0147	0.7946 ± 0.0863	0.8994 ± 0.0114	0.9036 ± 0.0596
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	0.8436 ± 0.0590	0.8690 ± 0.0110	0.8850 ± 0.0192	0.8808 ± 0.0341
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	0.8653 ± 0.0167	0.8992 ± 0.0156
	F_PP	0.8869 ± 0.0265	0.8643 ± 0.0275	0.8241 ± 0.0286	0.8744 ± 0.0192
	I_PP	0.9219 ± 0.0138	0.8932 ± 0.0106	0.8532 ± 0.0180	0.8931 ± 0.0163
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	0.9070 ± 0.0136	0.8728 ± 0.0126	0.8350 ± 0.0189	0.8797 ± 0.0150
SV	FA_SV_noOpt	0.8900 ± 0.0220	-	0.9086 ± 0.0204	0.9263 ± 0.0213
	FA_SV_opt	0.9149 ± 0.0170	-	0.9179 ± 0.0214	0.9313 ± 0.0213
	F_SV	-	-	-	0.913
	I_SV	0.9129 ± 0.0169	0.9177 ± 0.0149	0.8976 ± 0.0242	0.9014 ± 0.0530
	TQA_SV_noOpt	0.8475 ± 0.0380	0.8204 ± 0.0492	0.7859 ± 0.0629	0.8436 ± 0.0498
	TQA_SV_opt	0.8943 ± 0.0192	0.8952 ± 0.0191	0.8740 ± 0.0248	0.9114 ± 0.0252
	TS_SV	0.9027 ± 0.0223	0.9140 ± 0.0167	0.8770 ± 0.0226	0.8921 ± 0.0364

Table 8: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

5 Tables for depth $P = 5$

graph_type depth		ER 5	HH 5	L2F 5	RR 5
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	10 [40]	31 [39-144]	130 [40-100]	21 [40-100]
	L_MPS	10 [40]	31 [39-144]	130 [40-100]	21 [40-100]
	RTS_MPS	-	1 [106]	-	2 [40-100]
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	20 [105-144]	10 [40]	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	10 [40]	31 [39-144]	130 [40-100]	21 [40-100]
	L_PP	10 [40]	31 [39-144]	130 [40-100]	21 [40-100]
	RTS_PP	-	1 [106]	-	2 [40-100]
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	20 [105-144]	10 [40]	-
SV	FA_SV_noOpt	39 [10-20]	-	-	63 [10-20]
	FA_SV_opt	39 [10-20]	-	-	63 [10-20]
	F_SV	-	-	-	1 [10]
	L_SV	82 [10-20]	10 [12]	-	166 [10-20]
	TQA_SV_noOpt	82 [10-20]	10 [12]	-	133 [10-20]
	TQA_SV_opt	81 [10-20]	10 [12]	-	132 [10-20]
	TS_SV	81 [10-20]	10 [12]	-	132 [10-20]

Table 9: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 5	HH 5	L2F 5	RR 5
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	0.7947 ± 0.0818	0.7705 ± 0.0890	0.8584 ± 0.0462	0.8929 ± 0.0819
	I_MPS	0.95371 ± 0.00817	0.8152 ± 0.0873	0.9236 ± 0.0108	0.9321 ± 0.0363
	RTS_MPS	-	0.711	-	0.8373 ± 0.0729
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	0.7516 ± 0.0484	0.91392 ± 0.00952	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	0.9094 ± 0.0135	0.8842 ± 0.0265	0.8354 ± 0.0254	0.8898 ± 0.0153
	I_PP	0.9350 ± 0.0138	0.91053 ± 0.00968	0.8664 ± 0.0178	0.9054 ± 0.0178
	RTS_PP	-	0.882	-	0.8897 ± 0.0226
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	0.88187 ± 0.00789	0.8354 ± 0.0196	-
SV	FA_SV_noOpt	0.9001 ± 0.0226	-	-	0.9468 ± 0.0178
	FA_SV_opt	0.9318 ± 0.0151	-	-	0.9517 ± 0.0181
	F_SV	-	-	-	0.931
	I_SV	0.9300 ± 0.0146	0.92451 ± 0.00869	-	0.9160 ± 0.0494
	TQA_SV_noOpt	0.8685 ± 0.0353	0.8372 ± 0.0489	-	0.8682 ± 0.0463
	TQA_SV_opt	0.9159 ± 0.0152	0.9071 ± 0.0202	-	0.9317 ± 0.0219
	TS_SV	0.9188 ± 0.0208	0.9262 ± 0.0132	-	0.9034 ± 0.0351

Table 10: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

6 Tables for depth $P = 6$

	graph_type depth	ER 6	HH 6	L2F 6	RR 6
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	10 [100]	10 [40]
	F_MPS	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	I_MPS	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	10 [40]	10 [39]	130 [40–100]	20 [40]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	10 [100]	10 [40]
	F_PP	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	I_PP	10 [40]	31 [39–144]	130 [40–100]	21 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	10 [40]	10 [39]	130 [40–100]	20 [40]
SV	FA_SV_noOpt	16 [11–17]	-	-	43 [10–20]
	FA_SV_opt	16 [11–17]	-	-	43 [10–20]
	F_SV	-	-	-	-
	LSV	82 [10–20]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	-	133 [10–20]
	TQA_SV_opt	81 [10–20]	10 [12]	-	132 [10–20]
	TS_SV	80 [10–20]	10 [12]	-	132 [10–20]

Table 11: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 6	HH 6	L2F 6	RR 6
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	0.9514 ± 0.0108	0.96861 ± 0.00579
	F_MPS	0.8072 ± 0.0736	0.7659 ± 0.0998	0.8791 ± 0.0353	0.9219 ± 0.0671
	I_MPS	0.96470 ± 0.00899	0.8188 ± 0.0968	0.9386 ± 0.0105	0.9435 ± 0.0358
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	0.9120 ± 0.0290	0.88806 ± 0.00875	0.9115 ± 0.0135	0.9394 ± 0.0129
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	0.8908 ± 0.0181	0.9252 ± 0.0155
	F_PP	0.9204 ± 0.0150	0.8931 ± 0.0280	0.8475 ± 0.0217	0.8974 ± 0.0183
	I_PP	0.9437 ± 0.0139	0.92244 ± 0.00938	0.8757 ± 0.0176	0.9156 ± 0.0185
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	0.9327 ± 0.0155	0.89764 ± 0.00956	0.8516 ± 0.0217	0.9066 ± 0.0156
SV	FA_SV_noOpt	0.9049 ± 0.0249	-	-	0.9581 ± 0.0133
	FA_SV_opt	0.9447 ± 0.0150	-	-	0.9620 ± 0.0140
	F_SV	-	-	-	-
	I_SV	0.9422 ± 0.0131	0.9360 ± 0.0135	-	0.9220 ± 0.0539
	TQA_SV_noOpt	0.8825 ± 0.0331	0.8559 ± 0.0439	-	0.8841 ± 0.0440
	TQA_SV_opt	0.9296 ± 0.0143	0.9124 ± 0.0220	-	0.9429 ± 0.0216
	TS_SV	0.9306 ± 0.0195	0.9314 ± 0.0129	-	0.9141 ± 0.0345

Table 12: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

7 Tables for depth $P = 7$

graph_type depth		ER 7	HH 7	L2F 7	RR 7
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	I_MPS	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	I_PP	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	16 [11–17]	-	-	42 [10–20]
	FA_SV_opt	16 [11–17]	-	-	42 [10–20]
	F_SV	-	-	-	-
	I_SV	82 [10–20]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	-	132 [10–20]
	TQA_SV_opt	81 [10–20]	10 [12]	-	131 [10–20]
	TS_SV	80 [10–20]	10 [12]	-	130 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 7	HH 7	L2F 7	RR 7
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	0.8170 ± 0.0738	0.776 ± 0.101	0.8809 ± 0.0411	0.9102 ± 0.0756
	I_MPS	0.97193 ± 0.00935	0.8374 ± 0.0935	0.9486 ± 0.0100	0.9510 ± 0.0388
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	0.9031 ± 0.0384	0.8987 ± 0.0270	0.8519 ± 0.0207	0.9011 ± 0.0184
	I_PP	0.9491 ± 0.0139	0.93143 ± 0.00901	0.8826 ± 0.0175	0.9153 ± 0.0196
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	0.9101 ± 0.0295	-	-	0.9671 ± 0.0117
	FA_SV_opt	0.9550 ± 0.0156	-	-	0.9717 ± 0.0125
	F_SV	-	-	-	-
	I_SV	0.9514 ± 0.0125	0.9425 ± 0.0125	-	0.9229 ± 0.0595
	TQA_SV_noOpt	0.8936 ± 0.0327	0.8698 ± 0.0411	-	0.8952 ± 0.0426
	TQA_SV_opt	0.9391 ± 0.0135	0.9177 ± 0.0224	-	0.9514 ± 0.0178
	TS_SV	0.9394 ± 0.0190	0.9384 ± 0.0104	-	0.9221 ± 0.0339

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

8 Tables for depth $P = 8$

	graph_type depth	ER 8	HH 8	L2F 8	RR 8
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	10 [100]	10 [40]
	F_MPS	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	L_MPS	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	10 [40]	10 [39]	120 [40–100]	20 [40]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	10 [100]	10 [40]
	F_PP	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	L_PP	10 [40]	31 [39–144]	120 [40–100]	21 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	10 [40]	10 [39]	120 [40–100]	20 [40]
SV	FA_SV_noOpt	16 [11–17]	-	-	43 [10–20]
	FA_SV_opt	16 [11–17]	-	-	43 [10–20]
	F_SV	-	-	-	-
	L_SV	82 [10–20]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	-	133 [10–20]
	TQA_SV_opt	81 [10–20]	10 [12]	-	132 [10–20]
	TS_SV	80 [10–20]	10 [12]	-	131 [10–20]

Table 15: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 8	HH 8	L2F 8	RR 8
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	$0.967\,36 \pm 0.008\,41$	$0.982\,88 \pm 0.002\,87$
	F_MPS	0.8165 ± 0.0404	0.768 ± 0.102	0.8821 ± 0.0516	0.9383 ± 0.0627
	I_MPS	0.9761 ± 0.0103	0.843 ± 0.101	$0.956\,23 \pm 0.009\,61$	0.9606 ± 0.0320
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	0.9380 ± 0.0283	$0.903\,13 \pm 0.009\,80$	0.9221 ± 0.0125	$0.964\,15 \pm 0.009\,92$
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	0.9034 ± 0.0186	0.9341 ± 0.0157
	F_PP	0.9182 ± 0.0226	0.9011 ± 0.0304	0.8566 ± 0.0201	0.9044 ± 0.0218
	I_PP	0.9529 ± 0.0137	$0.938\,73 \pm 0.008\,54$	0.8884 ± 0.0176	0.9185 ± 0.0211
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	0.9363 ± 0.0148	$0.911\,34 \pm 0.007\,20$	0.8564 ± 0.0246	0.9197 ± 0.0146
SV	FA_SV_noOpt	0.9179 ± 0.0316	-	-	0.9733 ± 0.0108
	FA_SV_opt	0.9641 ± 0.0141	-	-	0.9790 ± 0.0116
	F_SV	-	-	-	-
	I_SV	0.9581 ± 0.0119	0.9470 ± 0.0111	-	0.9261 ± 0.0601
	TQA_SV_noOpt	0.9028 ± 0.0339	0.8811 ± 0.0401	-	0.9047 ± 0.0434
	TQA_SV_opt	0.9444 ± 0.0132	0.9234 ± 0.0184	-	0.9570 ± 0.0166
	TS_SV	0.9462 ± 0.0183	$0.943\,00 \pm 0.009\,62$	-	0.9285 ± 0.0334

Table 16: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

9 Tables for depth $P = 9$

	graph_type depth	ER 9	HH 9	L2F 9	RR 9
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	1 [40]	31 [39–144]	111 [40–100]	11 [40–100]
	I_MPS	1 [40]	31 [39–144]	111 [40–100]	11 [40–100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	1 [40]	31 [39–144]	111 [40–100]	11 [40–100]
	I_PP	1 [40]	31 [39–144]	111 [40–100]	11 [40–100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	16 [11–17]	-	-	42 [10–20]
	FA_SV_opt	16 [11–17]	-	-	42 [10–20]
	F_SV	-	-	-	-
	LSV	82 [10–20]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt	82 [10–20]	10 [12]	-	132 [10–20]
	TQA_SV_opt	81 [10–20]	10 [12]	-	131 [10–20]
	TS_SV	80 [10–20]	10 [12]	-	130 [10–20]

Table 17: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 9	HH 9	L2F 9	RR 9
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	-	-
	F_MPS	0.780	0.7833 ± 0.0972	0.8863 ± 0.0432	0.9268 ± 0.0735
	I_MPS	0.986	0.8618 ± 0.0927	0.96178 ± 0.00929	0.9308 ± 0.0449
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	-	-	-	-
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	-	-
	F_PP	0.908	0.9064 ± 0.0318	0.8617 ± 0.0195	0.9104 ± 0.0261
	I_PP	0.964	0.94445 ± 0.00803	0.8920 ± 0.0166	0.9078 ± 0.0226
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	-	-	-	-
SV	FA_SV_noOpt	0.9208 ± 0.0335	-	-	0.9774 ± 0.0105
	FA_SV_opt	0.9699 ± 0.0145	-	-	0.9836 ± 0.0108
	F_SV	-	-	-	-
	I_SV	0.9627 ± 0.0112	0.9478 ± 0.0124	-	0.9246 ± 0.0667
	TQA_SV_noOpt	0.9097 ± 0.0357	0.8905 ± 0.0377	-	0.9124 ± 0.0451
	TQA_SV_opt	0.9498 ± 0.0131	0.9296 ± 0.0161	-	0.9620 ± 0.0170
	TS_SV	0.9504 ± 0.0186	0.94662 ± 0.00897	-	0.9313 ± 0.0326

Table 18: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

10 Tables for depth $P = 10$

	graph_type depth	ER 10	HH 10	L2F 10	RR 10
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	10 [100]	11 [40-100]
	F_MPS	1 [40]	31 [39-144]	111 [40-100]	11 [40-100]
	I_MPS	1 [40]	31 [39-144]	111 [40-100]	11 [40-100]
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	1 [40]	31 [39-144]	111 [40-100]	11 [40-100]
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	10 [100]	11 [40-100]
	F_PP	1 [40]	31 [39-144]	111 [40-100]	11 [40-100]
	I_PP	1 [40]	31 [39-144]	111 [40-100]	11 [40-100]
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	1 [40]	30 [39-144]	111 [40-100]	11 [40-100]
SV	FA_SV_noOpt	16 [11-17]	-	-	26 [10-24]
	FA_SV_opt	16 [11-17]	-	-	25 [10-22]
	F_SV	-	-	-	-
	LSV	82 [10-20]	10 [12]	-	140 [10-20]
	TQA_SV_noOpt	82 [10-20]	10 [12]	-	141 [10-22]
	TQA_SV_opt	81 [10-20]	9 [12]	-	139 [10-20]
	TS_SV	79 [10-20]	9 [12]	-	136 [10-19]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 10	HH 10	L2F 10	RR 10
MPS	FA_MPS_no_opt	-	-	-	-
	FA_MPS_opt	-	-	$0.975\,93 \pm 0.007\,21$	$0.986\,01 \pm 0.009\,18$
	F_MPS	0.828	0.774 ± 0.109	0.8868 ± 0.0564	0.9280 ± 0.0900
	I_MPS	0.988	0.866 ± 0.101	$0.966\,26 \pm 0.008\,90$	0.9383 ± 0.0455
	RTS_MPS	-	-	-	-
	TQA_MPS_no_opt	-	-	-	-
	TQA_MPS_opt	0.952	0.8075 ± 0.0930	0.9260 ± 0.0165	0.9655 ± 0.0173
PP	FA_PP_no_opt	-	-	-	-
	FA_PP_opt	-	-	0.9067 ± 0.0188	0.9364 ± 0.0158
	F_PP	0.891	0.9110 ± 0.0325	0.8599 ± 0.0222	0.9127 ± 0.0258
	I_PP	0.966	$0.949\,10 \pm 0.007\,64$	0.8961 ± 0.0167	0.9011 ± 0.0244
	RTS_PP	-	-	-	-
	TQA_PP_no_opt	-	-	-	-
	TQA_PP_opt	0.949	$0.917\,75 \pm 0.006\,73$	0.8557 ± 0.0299	0.9225 ± 0.0158
SV	FA_SV_noOpt	0.9237 ± 0.0346	-	-	$0.978\,18 \pm 0.009\,62$
	FA_SV_opt	0.9740 ± 0.0141	-	-	$0.985\,20 \pm 0.009\,14$
	F_SV	-	-	-	-
	I_SV	0.9661 ± 0.0106	0.9484 ± 0.0111	-	0.9399 ± 0.0605
	TQA_SV_noOpt	0.9146 ± 0.0375	0.8988 ± 0.0356	-	0.9135 ± 0.0486
	TQA_SV_opt	0.9543 ± 0.0136	0.9322 ± 0.0206	-	0.9679 ± 0.0150
	TS_SV	0.9528 ± 0.0199	$0.951\,90 \pm 0.008\,32$	-	0.9426 ± 0.0314

Table 20: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).